Input: $y \in \{0,1\}^{m_i}$

Output: $a \in \{0,1\}$

Initialize $v \leftarrow$ root of $\boldsymbol{T}$
while v is not a leaf **do**
    if $i \in I^v$ **then**
        Let w be any vector in $(\mathrm{col}(Q^{\preceq v}) \setminus \mathrm{col}(Q^{\prec v})) \cap W_i$, query $\langle y, w \rangle$
        Compute $b^v \coloneqq \langle y, Q^v \rangle$ from $b^{\prec v}$ and $\langle y, w \rangle$
        Move $v \leftarrow \mathrm{child}(v, b^v)$
    else
        Sample $\boldsymbol{\xi} \sim \mathsf{Ber}(1/2)$
        Move $v \leftarrow \mathrm{child}(v, \boldsymbol{\xi})$
    end if
end while
return $L_i(v)$